# Tanmay Bhuskute

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## SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with 2 years of production experience building reliable, scalable backend services, full-stack apps, and cloud platforms using Python, Java, SQL, React, AWS, and Kubernetes. Current research in LLM-powered Android testing.

## EDUCATION

**Master of Science in Computer Science** Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

**Bachelor of Engineering in Computer Engineering** Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

## TECHNICAL SKILLS

**Programming:** Python, Java, SQL, JavaScript, TypeScript, C/C++, Go, Data Structures & Algorithms.

**Web & Backend:** React, Node.js, FastAPI, Flask, Django, HTML/CSS, REST APIs, Microservices.

**Databases:** PostgreSQL, MySQL, DynamoDB, Redis.

**Developer Tools:** Git, GitLab CI/CD, AWS, Docker, Kubernetes, Terraform, Linux, SonarQube, CloudWatch, Datadog, Pytest, JUnit.

**AI & Data:** Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

## WORK EXPERIENCE

**Graduate Researcher** | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

**Software Development Engineer** | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** workflows with Python (Boto3), replacing legacy Ruby scripts, cutting AWS deployment time **40%**.
- Executed **44 production environment upgrades** in 4 months, diagnosing live issues across Linux, EC2, databases, and services using CloudWatch and Datadog monitoring.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** and enabling reliable rollback during live deployments.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%** via coding standards and CI/CD quality gates; mentored **5 software engineers**.

**Software Developer Intern** | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** tests in pre-deployment workflow, catching approximately **95%** of historical misconfigurations.

**Software Developer Intern** | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built full-stack **React and MySQL** records application and deployed on-premises database with authentication and user permissions, cutting document retrieval time by **90%** (5 min to 30 sec).

## PROJECTS AND RESEARCH

**NYC Mobility Forecasting** [Python, FastAPI, PostgreSQL, Redis, PySpark] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** dataset), cutting RMSE **43%** over baseline; served real-time predictions via FastAPI with PostgreSQL and Redis caching.

**Spendora: AI Personal Finance Platform** [Next.js, TS, PostgreSQL, Supabase, Gemini] Jun 2025

- Architected **PostgreSQL** financial ledger supporting refunds, EMIs, and **120+ currencies**, validating cent-level reconciliation accuracy across **18 months** of historical data.
- Benchmarked Gemini-powered assistant on 100 queries (**94%** numerical correctness, 2.1s median latency) and shipped **80 automated tests** across unit, database, and end-to-end layers.

## EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).